Homework-4

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6).

(a) Hidden layer computations

$$z_j = w_{1j}x_1 + w_{2j}x_2 + w_{3j}x_3 + b_j$$

$$h_j = \sigma(z_j) = \frac{1}{1+e^{-z_j}}$$

since input's

$x_1 = 2, x_2 = 1, x_3 = 3.$

z_1

$$z_1 = 0.2(2) + 0.1(1) + (-0.2)(3) + 0.4$$

$$z_1 = 0.4 + 0.1 - 0.6 + 0.4 = 0.3$$

z_2

$$z_2 = (-0.5)(2) + (0.3)(1) + (0.8)(3) - 0.6$$

$$= -1 + 0.3 + 2.4 - 0.6 = 1.1$$

sigmoid

$$h_1 = \frac{1}{1+e^{-0.3}} = 0.574$$

$$h_2 = \frac{1}{1+e^{-1.1}} = 0.750$$

). pre activation:

$$z = 0.7(0.574) + (-1.2)(0.750) + 0.5$$

$$z = 0.4018 - 0.900 + 0.5$$

$$= 0.0018$$

<u>sigmoid o/p</u>

$$y = \frac{1}{1+e^{-0.0018}} = 0.50045$$

①. Given, $t = 1$

$$L = -\log(y)$$

$$L = -\log(0.50045)$$

$$= 0.6927$$

). Given, $h_1 = ReLU(x_1 + x_2)$, $h_2 = ReLU(x_1 + x_2 - 1)$

$h_3 = ReLU(2x_1 - x_2)$, $y = ReLU(h_1 - 2h_2 + h_3)$

). (0,0)

$h_1 = ReLU(0+0) = 0$

$h_2 = ReLU(0+0-1) = 0$

$h_3 = ReLU(0-0) = 0$

$y = ReLU(0-0+0) = 0$

(0,1)

$h_1 = ReLU(1) = 1$

$h_2 = ReLU(0) = 0$

$h_3 = ReLU(-1) = 0$

$y = ReLU(1-0+0) = 1$

(1,0)

$h_1 = ReLU(1) = 1$

$h_2 = ReLU(0) = 0$

$h_3 = ReLU(2) = 2$

$y = ReLU(1-0+2) = 3$

(1,1)

$h_1 = ReLU(2) = 2$

$h_2 = ReLU(1) = 1$

$h_3 = ReLU(1) = 1$

$y = ReLU(2-2+1) = 1$

(b). Decision Boundary Comparison

⇒ original XOR Network:

* 2 hidden units
* simpler piecewise linear boundary

⇒ New network:

* 3 hidden units
* more complex & flexible boundary

(c). XOR correctness

(0,0) = 0, (0,1) = 1, (1,0) = 1, (1,1) = 0

Input	output	XOR	Match
(0,0)	0	0	✓
(0,1)	1	1	✓
(1,0)	3	1	✗
(1,1)	1	0	✗

Therefore,

⇒ The network doesnot compute XOR exactly

⇒ Incorrect for: (1,0) (1,1).

(8).

(a). Decision Boundary

$$w_1 x_1 + w_2 x_2 + b = 0$$

$$x_1 - 2x_2 + 1 = 0$$

(b). compute

$$z = x_1 - 2x_2 + 1$$

Points	z	Prediction	True	Result
(2,1)	1	1	1	✓
(1,3)	-4	0	0	✓
(3,2)	0	0	1	✗
(0,1)	-1	0	0	✓

⇒ (3,2) is misclassified point.

(c) Perceptron Loss

No. of mistakes = 1.

(d) weight update

$$w = w + \eta \cdot (t - y) \cdot x$$

$$b = b + \eta \cdot (t - y)$$

<u>weights</u>

$\Rightarrow w_1 = 1 + 0.5(1)(3) = 1 + 1.5 = 2.5$

$w_2 = -2 + 0.5(1)(2) = -2 + 1 = -1$

<u>bias</u>

$b = 1 + 0.5(1) = 1.5$